

GELA, Italy

WMO: 164530

Lat: 37.076N

Lon: 14.225E

Elev: 226

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	42.0	44.4	30.0	24.3	48.9	33.4	28.2	49.0	33.8	56.8	30.1	56.4	5.6	30	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	85.4	76.0	83.5	75.9	82.0	75.6	79.6	83.1	78.3	81.9	77.0	80.9	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	149.6	82.6	77.0	141.9	81.5	75.6	135.2	80.4	43.4	83.2	41.8	82.2	40.6	81.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.6	27.2	23.0	37.9	90.9	3.4	2.9	35.5	93.0	33.5	94.7	31.6	96.3	29.1	98.4		
(5)			35.8	81.7	2.8	1.8	33.8	83.0	32.2	84.0	30.6	85.0	28.6	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	54.3	53.5	54.7	58.1	64.2	70.6	74.7	77.4	74.0	68.6	61.3	56.2	(6)
(7)		DBStd	9.25	3.33	3.53	4.09	3.84	4.49	3.96	3.70	2.86	3.30	3.89	4.57	3.92	(7)
(8)		HDD50	32	6	9	11	0	0	0	0	0	0	0	1	5	(8)
(9)		HDD65	1674	332	321	320	211	71	3	1	0	0	14	126	275	(9)
(10)		CDD50	5150	139	109	156	242	439	618	766	848	720	577	340	196	(10)
(11)		CDD65	1315	0	0	0	4	44	171	302	383	270	125	15	1	(11)
(12)		CDH74	6858	0	0	0	3	89	466	1586	3052	1415	244	2	1	(12)
(13)		CDH80	863	0	0	0	0	7	43	202	489	115	7	0	0	(13)
(14)	Wind	WSAvg	8.5	7.9	8.7	9.2	10.5	9.0	8.8	7.7	7.6	8.0	7.8	8.5	8.3	(14)
(15)	Precipitation	PrecAvg	21.2	3.1	2.2	1.9	1.4	0.6	0.4	0.2	0.4	2.0	2.5	3.0	3.4	(15)
(16)		PrecMax	34.4	8.9	9.0	6.8	3.3	1.7	1.5	1.1	1.2	5.6	5.2	8.4	8.9	(16)
(17)		PrecMin	10.6	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.4	0.2	(17)
(18)		PrecStd	5.9	2.1	2.0	1.5	0.9	0.5	0.4	0.2	0.4	1.7	1.4	1.9	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.3	65.5	68.7	73.0	80.5	84.3	87.6	88.8	85.4	82.1	73.7	68.0	(19)
(20)			MCWB	59.0	56.7	56.7	59.4	64.9	68.6	74.1	76.9	72.8	74.2	67.2	61.6	(20)
(21)		2%	DB	62.7	62.6	64.8	69.5	76.6	80.4	84.6	85.8	82.6	78.5	71.7	65.5	(21)
(22)			MCWB	57.1	56.2	56.8	59.9	65.8	68.7	75.4	77.9	75.1	71.6	65.6	60.0	(22)
(23)		5%	DB	61.1	61.0	62.7	66.5	73.7	77.9	82.6	84.0	80.9	76.8	69.9	64.0	(23)
(24)			MCWB	56.0	55.7	56.7	59.4	65.3	70.0	75.3	77.5	74.7	71.1	64.4	59.0	(24)
(25)		10%	DB	59.8	59.8	61.1	64.3	71.4	76.5	80.7	82.5	79.7	75.3	68.2	62.3	(25)
(26)			MCWB	55.0	54.9	56.1	59.3	64.8	70.0	74.0	76.9	74.2	70.3	63.0	57.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	59.8	60.8	63.2	71.8	75.5	81.0	81.9	78.8	76.4	69.5	63.4	(27)
(28)			MCD B	63.6	62.7	64.4	67.3	75.5	78.8	83.9	84.9	82.1	79.4	72.6	66.8	(28)
(29)		2%	WB	58.6	58.2	59.2	61.9	69.3	73.7	78.8	80.4	77.4	73.8	66.9	61.3	(29)
(30)			MCD B	61.8	61.1	62.6	65.9	72.9	76.9	82.5	83.7	80.7	76.6	70.5	64.6	(30)
(31)		5%	WB	57.1	57.0	57.9	60.8	67.4	72.4	77.0	79.0	76.3	72.3	65.1	59.6	(31)
(32)			MCD B	60.2	60.2	61.4	65.0	71.1	75.9	81.1	82.6	79.8	75.8	69.0	63.0	(32)
(33)		10%	WB	55.7	55.7	56.7	59.7	65.8	71.1	75.3	77.9	75.1	70.9	63.7	58.0	(33)
(34)			MCD B	59.0	59.2	60.4	63.7	69.9	75.0	79.3	81.7	79.0	74.8	67.7	61.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.6	11.1	10.5	9.7	8.5	8.3	8.5	9.6	10.1	10.1	9.9	(35)
(36)			MCD BR	10.5	11.8	12.9	12.9	12.6	11.2	10.7	10.5	11.0	11.1	9.6	11.3	(36)
(37)			MCWBR	7.8	9.0	9.9	8.7	10.2	10.3	11.1	11.4	10.8	8.8	7.5	8.5	(37)
(38)		5% WB	MCD BR	10.4	10.6	11.5	10.9	9.9	8.6	8.2	9.1	10.3	10.0	8.7	10.0	(38)
(39)			MCWBR	8.9	9.2	10.1	9.0	10.8	11.0	11.9	11.3	10.8	9.5	7.6	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.351	0.373	0.422	0.463	0.465	0.444	0.446	0.454	0.448	0.422	0.386	0.356	(40)	
(41)		taud	2.405	2.322	2.171	2.076	2.118	2.216	2.224	2.201	2.225	2.272	2.329	2.397	(41)	
(42)		Ebn at Noon	262	270	266	261	262	267	265	260	255	249	246	251	(42)	
(43)		Edh at Noon	28	35	44	51	50	45	44	44	41	36	30	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	750	1007	1417	1810	2159	2395	2461	2167	1641	1187	830	676	(44)	
(45)		RadStd	58	95	101	115	112	108	64	105	75	78	59	57	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.67	-0.85	N/A	N/A	N/A		

Nomenclature: See separate page